Software requirements specification

- Nupjuk Campus -

version 1.1

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1. Introduction

1.1 Purpose

This document specifies the software requirements for **“Nupjuk Campus”**, a web-based platform integrated with a Chrome extension that provides KAIST students with course-based community features, assignment/project scheduling, and team meeting coordination.

This SRS is intended for the development team who will implement this system.

1.2 Scope

Nupjuk Campus consists of two components

1. **Chrome Extension**: Accesses the KLMS web pages within the user’s active browser session to parse course-related data (enrolled courses, assignments, deadlines, announcements) and transmits it to the service backend.
2. **Web Service**: A standalone web application that provides course-based bulletin boards, integrated schedule management, and team meeting scheduling functionality.

The system aims to solve the fragmentation problem where KAIST students currently rely on multiple disconnected tools (KLMS for course content, Everytime for community, when2meet for scheduling) by unifying these experiences around course-based contexts.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
KLMS	KAIST Learning Management System (Moodle-based)
SSO	Single Sign-On, KAIST’s unified authentication system
Moodle	Open-source learning management platform on which KLMS is built
Chrome Extension	A browser add-on for Google Chrome that can interact with web page content

1.4 References

- IEEE 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- KLMS (https://klms.kaist.ac.kr)
- Moodle Developer Documentation (https://moodledev.io)
- Chrome Extensions Documentation (https://developer.chrome.com/docs/extensions)

1.5 Overview

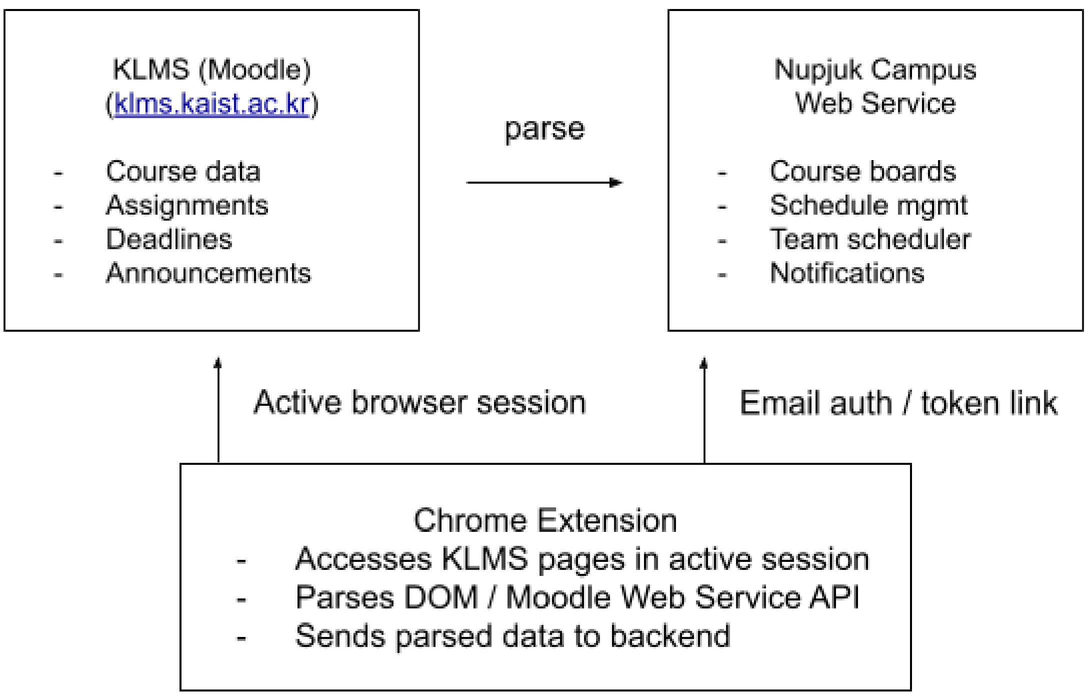
Section 2 provides an overall description of the product. Section 3 describes external interface requirements. Section 4 details specific functional and non-functional requirements.

2. Overall Description

2.1 Product Perspective

Nupjuk Campus is a new, self-contained system that bridges the gap between KLMS and students' daily academic workflow. It does not modify or replace KLMS but rather extends its utility by extracting data via a Chrome extension and presenting it within a richer, community-driven platform.

- **System Architecture Overview**



The Chrome extension does not perform SSO authentication itself; it simply accesses KLMS pages within the user's already-authenticated browser session to parse course data. The web service authenticates users independently through KAIST email registration and verification. Account linking between the extension and the web service is performed via a secure token exchange mechanism (see FR-AUTH-003).

2.2 Product Features (Summary)

- F1: Chrome extension for KLMS data parsing and synchronization
- F2: Course-based bulletin board / community
- F3: Integrated assignment and deadline schedule management
- F4: Team meeting scheduler with visual availability grid
- F5: Notification and reminder system
- F6: User account and authentication

2.3 User Classes and Characteristics

User Class	Description	Technical Proficiency
KAIST Student (Primary)	Undergraduate/graduate students enrolled in courses and using the service for course community and scheduling. A student's access level for each course board depends on whether their enrollment for that course has been verified through the Chrome extension (see Section 4.1.2 for access-control details).	Medium; familiar with web services
Team Leader	A student who creates and manages team scheduling events within a course context	Medium

2.4 Operating Environment

- **Chrome Extension**: Google Chrome browser (version 110+), Chromium-based browsers (Edge, Brave)

- **Web Service**: Modern web browsers (Chrome, Firefox, Safari, Edge); responsive design for mobile access
- **Server**: Cloud-hosted backend (specific technology at the discretion of the development team)

2.5 Design and Implementation Constraints

- The Chrome extension operates within the user's already-authenticated KLMS browser session to access and parse course data. The extension cannot and does not handle SSO credentials directly.
- Data parsing depends on KLMS's DOM structure or Moodle Web Service API; changes to KLMS may require extension updates.
- The system must comply with KAIST's data usage policies regarding student information.
- The Chrome extension must follow the Chrome Web Store's Manifest V3 requirements.

2.6 Assumptions and Dependencies

- Users have a valid KAIST email account and may register for the web service using that email address.
- Users who want to access course-specific boards or synced academic features must complete at least one successful KLMS sync through the Chrome extension during the relevant semester.
- Users use a Chrome-compatible browser to run the extension.
- KLMS remains Moodle-based and maintains its current structure (or Moodle Web Service APIs remain available).
- The service does not store SSO credentials. Web-service authentication is based on KAIST email registration and verification, while the Chrome extension is used to link the account and verify course enrollment data.
- Course-board access persists until the end of the semester (approximately 16 weeks from semester start) as recorded in the enrollment's valid_until field. If a subsequent sync indicates that the user is no longer enrolled in a course, access shall be revoked immediately regardless of the remaining valid_until

period. If no subsequent sync occurs, access is retained until valid_until expires, at which point it is automatically revoked.

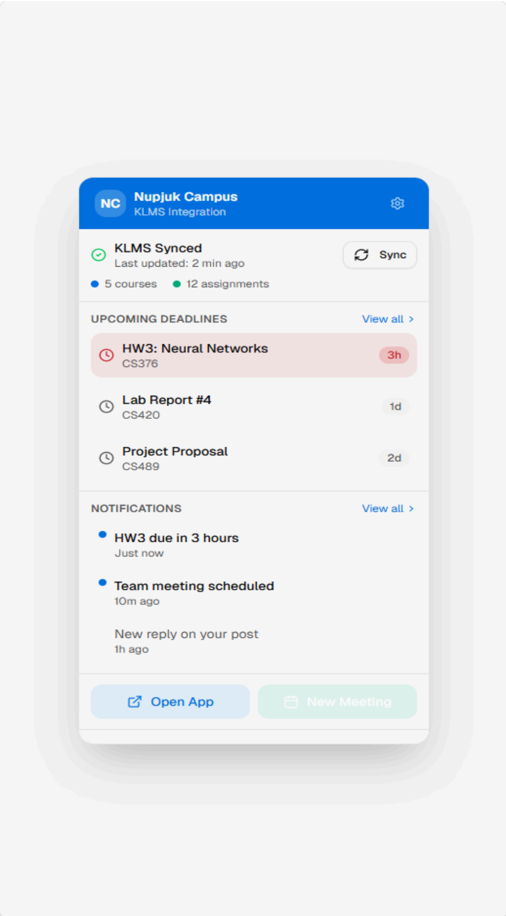
- The target user base is approximately 10,000 KAIST students (undergraduate and graduate combined).

3. External Interface Requirements

3.1 User Interfaces

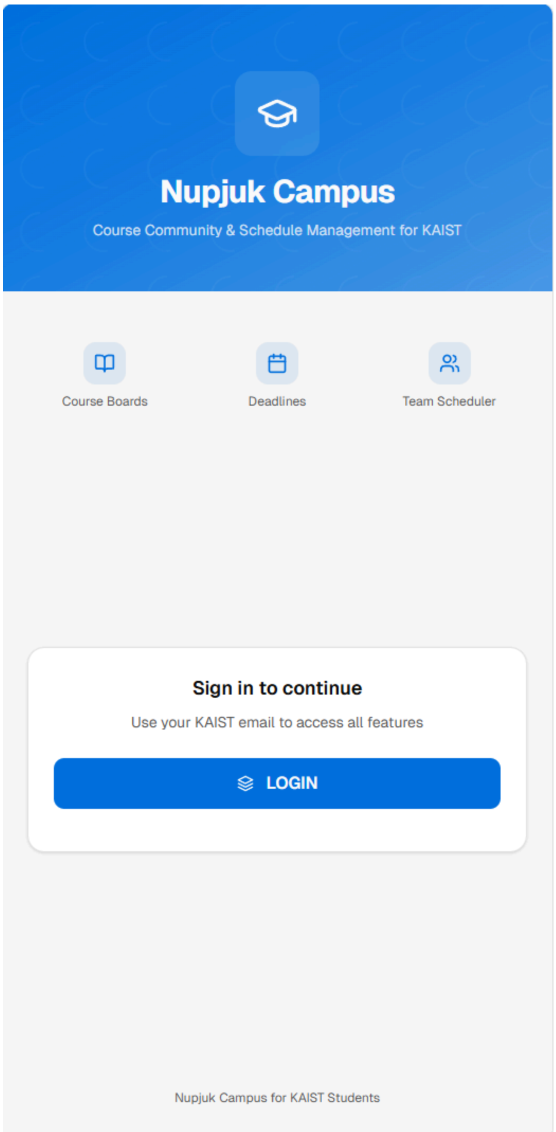
3.1.1 Chrome Extension Popup

- Displays current sync status and last sync time.
- Shows a list of synced courses with assignment counts.
- Provides a “Sync Now” button for manual synchronization.
- Provides a link to open the Nupjuk Campus web service.



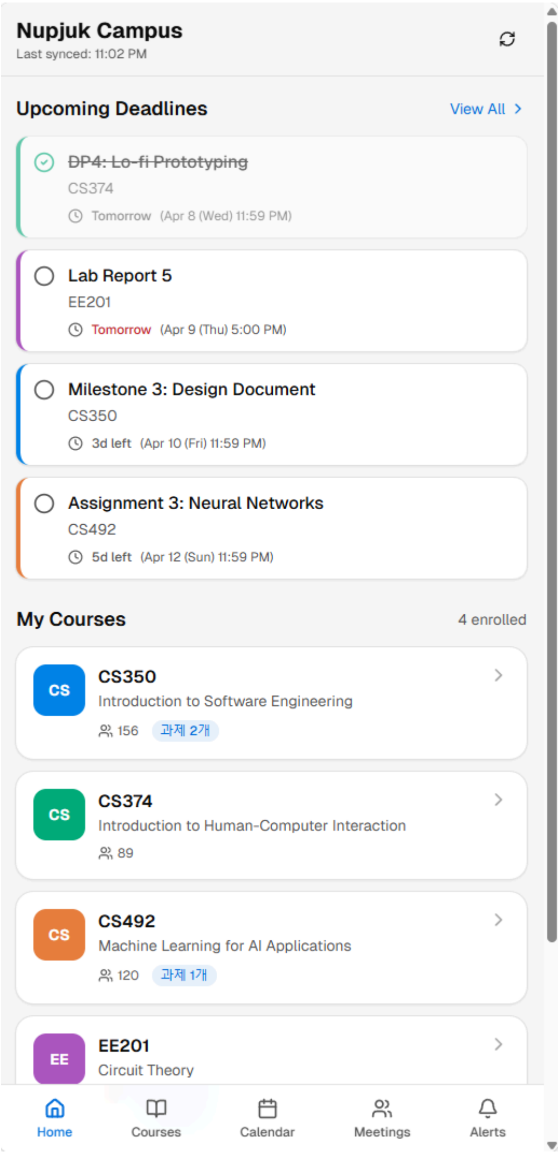
3.1.2 Web Service — Login

- Displays a login page for Nupjuk Campus users.
- Provides input fields for a KAIST email address and password, along with a "Log In" button.
- Provides a registration flow for new users using a KAIST email address.
- Sends a verification email during registration to confirm ownership of the KAIST email address.
- Displays an error message for invalid credentials, unsupported email domains, duplicate accounts, or unverified accounts.
- Navigates the user to the Main Dashboard upon successful login.



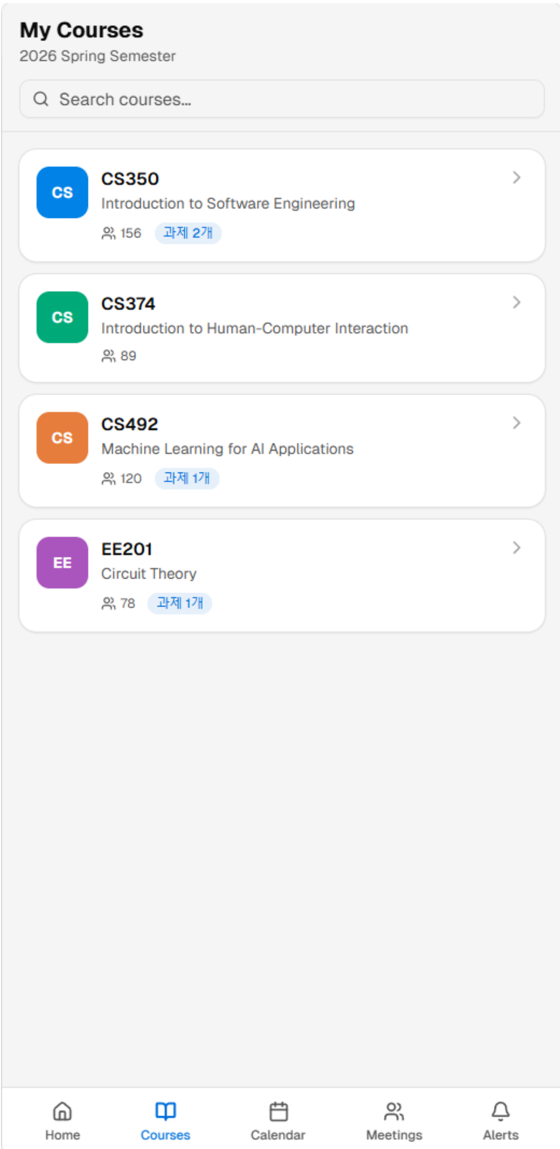
3.1.3 Web Service — Main Dashboard

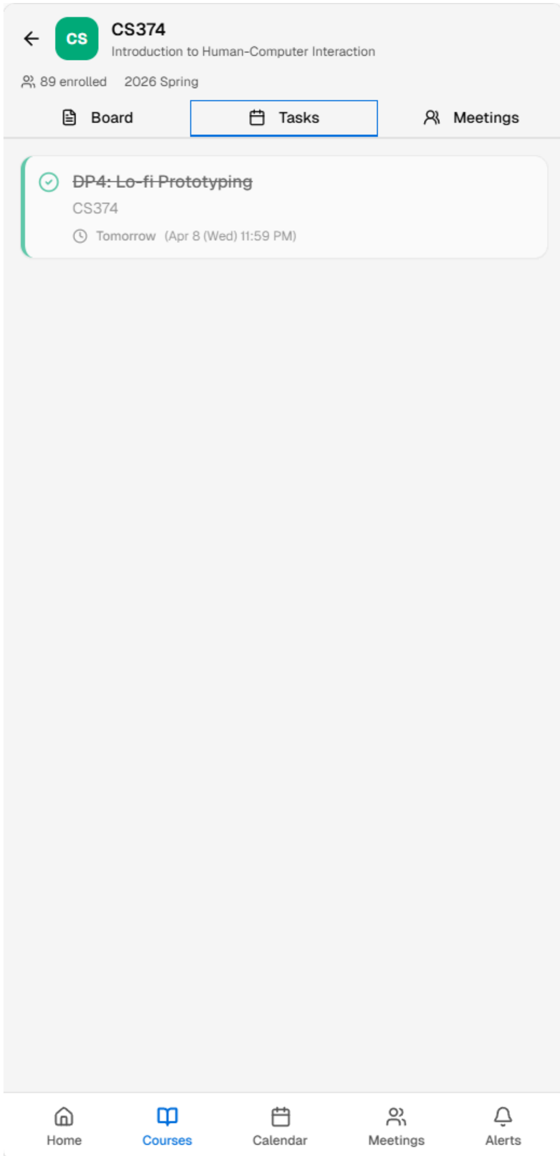
- Displays a list of enrolled courses whose enrollment has been verified through the linked extension during the current semester, with quick access to each course board.
- Shows upcoming deadlines in a summary widget (next 7 days).
- Shows recent notifications.
- Provides navigation to calendar view and meeting scheduler.



3.1.4 Web Service — Course Board

- Standard bulletin board layout with post list, post detail view, and comment section.
- Search bar at the top.
- An optional category filter may be provided in future versions.
- Users without verified enrollment may only see the board name and introduction or an access-restriction message.

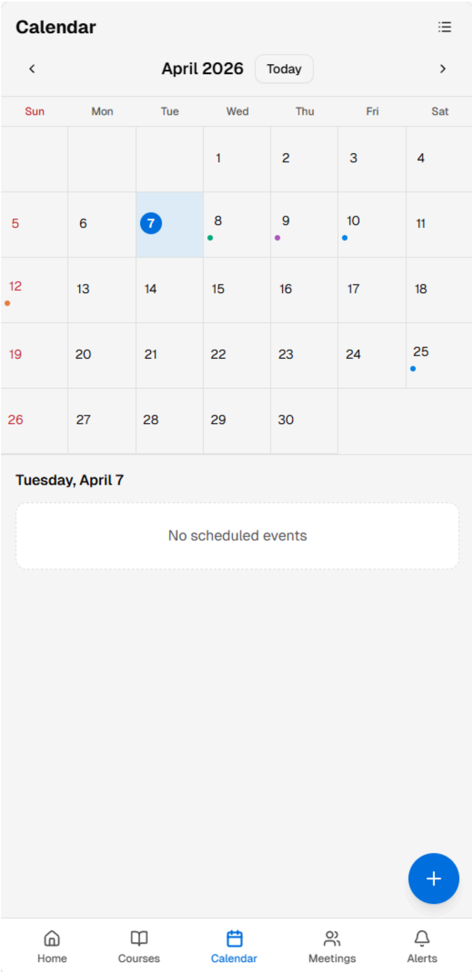




3.1.5 Web Service — Calendar View

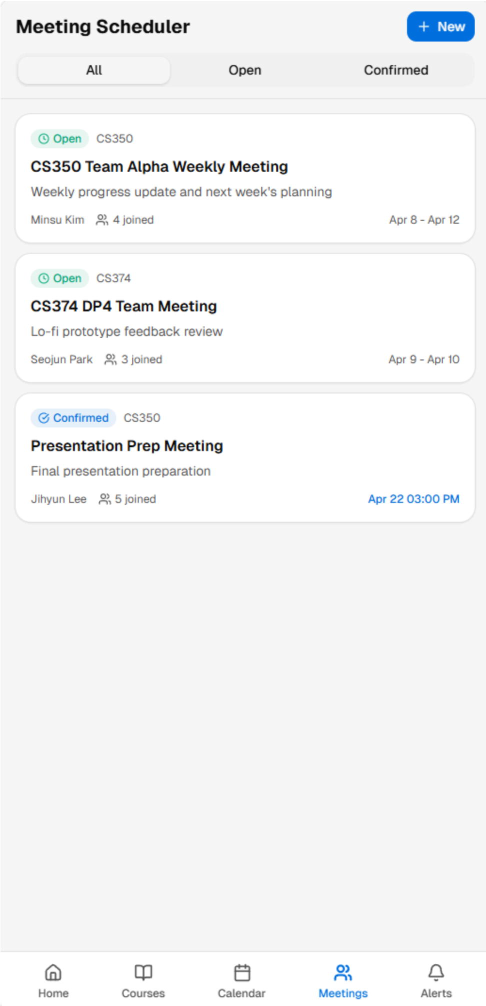
- Monthly/weekly/daily toggle for calendar display.
- Color-coded events per course.

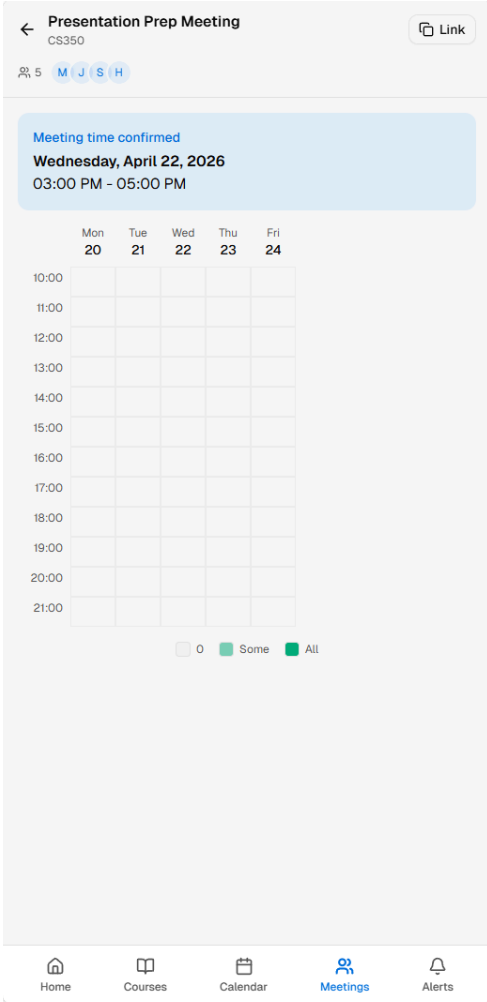
- Click-to-add functionality for personal events.
- Assignment items are clearly distinguished from personal events and meetings.



3.1.6 Web Service — Meeting Scheduler

- Visual time-slot grid with drag-to-select interaction.
- Real-time availability heatmap (color intensity indicates number of available participants) powered by WebSocket updates.
- Participant list with individual availability toggle.
- Mobile-friendly touch interaction for time slot selection.





3.2 Hardware Interfaces

No specific hardware interfaces are required.

3.3 Software Interfaces

Interface	Description
KLMS (Moodle)	The Chrome extension accesses the KLMS web pages within the user's active browser session to parse course and assignment data via DOM parsing or Moodle Web Service APIs. The extension does not perform SSO authentication; it relies on the user's already-authenticated session.
Backend API	RESTful API (or GraphQL, at the development team's discretion) for communication between the Chrome extension, web frontend, and server.

Interface	Description
Email Service	Integration with an email delivery service (e.g., SMTP, SendGrid) for email verification and optional email notifications.
File Storage Backend	Initial attachment storage is service-side server storage; the system shall be configurable to use S3-compatible object storage in future deployments.

3.4 Communication Interfaces

- HTTPS for all client-server communication.
- WebSocket for real-time updates on the meeting scheduler and in-app notifications.

4. Specific Requirements

4.1 Functional Requirements

4.1.1 Chrome Extension — KLMS Data Parsing (F1)

ID	Requirement	Priority
FR-EXT-001	The extension shall detect when the user has an active KLMS session in the browser (i.e., the user is logged into KLMS).	Must
FR-EXT-002	The extension shall parse the list of courses the user is currently enrolled in, including course name, course code, semester, and KLMS course ID (klms_course_id).	Must
FR-EXT-003	The extension shall parse assignment information for each enrolled course, including assignment title, due date/time, and submission status.	Must
FR-EXT-004	The extension shall parse course announcement/notice data including title, date, and content summary.	Should
FR-EXT-005	The extension shall transmit parsed data to the Nupjuk Campus backend via secure HTTPS API calls.	Must

ID	Requirement	Priority
FR-EXT-006	The extension shall perform data synchronization when the user visits KLMS, with an option for manual sync via the extension popup.	Must
FR-EXT-007	The extension shall display a badge or indicator showing the last sync time and sync status.	Should
FR-EXT-008	The extension shall handle parsing errors gracefully and notify the user if KLMS structure changes cause parsing failures.	Must
FR-EXT-009	The extension shall not store or transmit the user's SSO credentials.	Must
FR-EXT-010	The extension shall detect when a previously synced course is no longer present in the user's KLMS enrollment data and shall notify the backend so that course-specific access rights can be revoked immediately.	Must

4.1.2 Course-Based Bulletin Board / Community (F2)

ID	Requirement	Priority
FR-BRD-001	The system shall automatically create or map a bulletin board for each unique course identified by the combination of course_code, semester, and klms_course_id from extension sync data.	Must
FR-BRD-002	The system shall restrict full board access (viewing posts, creating posts, commenting, and participating in course-specific features) to users whose enrollment has been verified through at least one successful KLMS sync during the relevant semester.	Must
FR-BRD-003	Users shall be able to create posts on a course board with a title, body text, and optional file attachments.	Must
FR-BRD-004	Users shall be able to comment on posts.	Must
FR-BRD-005	Users shall be able to edit and delete posts/comments that they authored.	Must

ID	Requirement	Priority
FR-BRD-006	Users whose KAIST email is verified but whose enrollment for a course has not been verified shall be able to view limited board metadata, specifically the board name and introduction, but shall not be able to view posts or comments.	Should
FR-BRD-007	The system shall provide a search function within each course board (by title, content, author).	Should
FR-BRD-008	The system shall display the number of enrolled/active users on each course board.	Could
FR-BRD-009	Users shall be able to upvote/like posts and comments.	Could
FR-BRD-010	The system shall support categorization of posts (e.g., Q&A, Team Recruitment, General, Resources).	Should
FR-BRD-011	The system shall revoke access to a course board immediately when a subsequent KLMS sync indicates that the user is no longer enrolled in the corresponding course. If no subsequent sync occurs, the enrollment shall remain active until the valid_until date (end of semester), after which access is automatically revoked.	Must

4.1.3 Assignment and Schedule Management (F3)

ID	Requirement	Priority
FR-SCH-001	The system shall display all synced assignments and deadlines in a unified calendar view.	Must
FR-SCH-002	The system shall display assignments in a list view, sortable and filterable by course, due date, and completion status.	Must

ID	Requirement	Priority
FR-SCH-003	Users shall be able to manually add personal schedules or deadlines (e.g., exams, project milestones) to the calendar.	Must
FR-SCH-004	The system shall send reminders/notifications for upcoming deadlines (configurable: 1 day before, 3 hours before, etc.).	Should
FR-SCH-005	The system shall visually distinguish between different courses using color coding on the calendar.	Should
FR-SCH-006	The system shall allow users to mark assignments with a personal completion status independent of the synced KLMS submission status.	Must
FR-SCH-007	The system shall update synced assignment data when the extension performs a new sync, reflecting changes from KLMS without overwriting the user's personal completion status.	Must
FR-SCH-008	The system shall display both the synced KLMS submission status and the user's personal completion status where applicable.	Should

4.1.4 Team Meeting Scheduler (F4)

ID	Requirement	Priority
FR-MTG-001	Users shall be able to create a meeting scheduling event with a title, description, date range, and time range.	Must
FR-MTG-002	Users shall be able to share a meeting scheduling event via a unique link or within a course board; access to the meeting event shall be restricted to users whose enrollment has been verified for the corresponding course during the relevant semester.	Must

ID	Requirement	Priority
FR-MTG-003	Participants shall be able to indicate their available time slots on a visual grid interface.	Must
FR-MTG-004	The system shall display an aggregated availability heatmap showing overlapping free times of all participants.	Must
FR-MTG-005	The meeting creator shall be able to finalize a meeting time, which is then added to the internal service calendars of all participants.	Should
FR-MTG-006	The system shall automatically suggest the best meeting times based on maximum participant overlap.	Should
FR-MTG-007	The scheduling UI shall support drag-to-select time slots, mobile-friendly interaction, and real-time updates via WebSocket.	Must
FR-MTG-008	The system may pre-fill busy times in the scheduler using synced assignment/deadline data and user-created personal events stored within the service.	Could

4.1.5 Notification System (F5)

ID	Requirement	Priority
FR-NTF-001	The system shall automatically subscribe users to notifications for all course boards where their enrollment has been verified. Users cannot unsubscribe from a course board but may disable notifications for specific boards via notification preferences.	Should
FR-NTF-002	The system shall notify users of upcoming assignment deadlines based on synced data.	Must
FR-NTF-003	The system shall notify users of meeting scheduling requests and finalized meeting times.	Should

ID	Requirement	Priority
FR-NTF-004	Users shall be able to configure notification preferences, including enabling/disabling notifications per course board, per notification type, and setting reminder timing.	Should
FR-NTF-005	Notifications shall be delivered via an in-app notification center with real-time push via WebSocket, and may optionally be delivered via email depending on user preferences.	Should
FR-NTF-006	Each in-app notification shall include sufficient routing information to navigate the user to the related target resource, such as a board post, comment thread, assignment item, meeting event, or calendar entry.	Must

4.1.6 User Account and Authentication (F6)

ID	Requirement	Priority
FR-AUTH-001	The system shall allow users to register for the web service using a KAIST email address (@kaist.ac.kr).	Must
FR-AUTH-002	The system shall require email verification before activating a newly registered account.	Must
FR-AUTH-003	The system shall allow registered users to log in using their KAIST email address and password.	Must
FR-AUTH-004	The system shall link the Chrome extension to the user's authenticated web-service account via a secure token exchange. The flow is as follows: (1) the user logs in to the web service and obtains an authentication token; (2) the user authenticates the Chrome extension by entering or pasting this token, or via an automated browser-based handshake; (3) the extension stores the token locally and includes it in subsequent API calls to the backend for enrollment verification and data synchronization.	Must

ID	Requirement	Priority
FR-AUTH-005	Users shall be able to set a display name and optional profile information.	Should
FR-AUTH-006	The system shall provide session management with automatic logout after a configurable inactivity period.	Should
FR-AUTH-007	Course-specific privileges shall be granted based on enrollment verification obtained through the linked Chrome extension, not by email-domain verification alone.	Must

4.2 Non-Functional Requirements

4.2.1 Performance

ID	Requirement
NFR-PERF-001	Web pages shall load within 3 seconds when up to 500 concurrent users are connected under standard broadband network conditions (10 Mbps+).
NFR-PERF-002	The Chrome extension data sync shall complete within 30 seconds for a typical course load (5-8 courses).
NFR-PERF-003	The meeting scheduler heatmap shall update in real time (within 2 seconds) when a participant changes availability.

4.2.2 Security

ID	Requirement
NFR-SEC-001	All communication between the extension, web service, and backend shall use HTTPS/TLS encryption.
NFR-SEC-002	The system shall never store, transmit, or have access to users’ KAIST SSO passwords.

ID	Requirement
NFR-SEC-003	API endpoints shall require authentication tokens; unauthenticated requests shall be rejected.
NFR-SEC-004	User data shall be stored in compliance with relevant privacy regulations and KAIST data policies.

4.2.3 Usability

ID	Requirement
NFR-USE-001	The web service shall be accessible on both desktop and mobile browsers with responsive design.
NFR-USE-002	A new user shall be able to install the extension, sync data, and access a course board within 5 minutes.
NFR-USE-003	The UI shall support both Korean and English to accommodate international students.

4.2.4 Reliability and Availability

ID	Requirement
NFR-REL-001	The web service shall target 99% uptime per academic semester (approximately 16 weeks), measured as total available minutes divided by total minutes in the semester.
NFR-REL-002	The system shall handle KLMS structure changes gracefully, with informative error messages rather than silent failures.

4.2.5 Scalability

ID	Requirement
NFR-SCL-001	The system shall support at least 500 concurrent users during peak usage (e.g., exam periods), with a total

ID	Requirement
	registered user base of up to 10,000 students (undergraduate and graduate combined).
NFR-SCL-002	The system shall handle at least 200 active course boards simultaneously.

4.2.6 File Storage and Retention

ID	Requirement
NFR-FILE-001	File attachments uploaded to board posts shall initially be stored on the service-side server storage.
NFR-FILE-002	The system shall support configurable file-retention periods for uploaded attachments and shall automatically delete or expire files after the configured validity period.
NFR-FILE-003	The system shall enforce a maximum file size of 15 MB per post (total size of all attachments in a single post).
NFR-FILE-004	The system shall restrict uploaded file types to document and image formats only. Permitted extensions include: pdf, doc, docx, xls, xlsx, ppt, pptx, txt, hwp, hwpx, csv, jpg, jpeg, png, gif, bmp, webp. All other file types, including executable files and SVG, shall be rejected.
NFR-FILE-005	The attachment storage layer shall be designed so that the storage backend can be switched through configuration, including support for future migration to S3-compatible object storage.
NFR-FILE-006	If SVG support is added in a future version, SVG files shall be sanitized to remove embedded scripts, external references, and potentially harmful content before storage, and shall be served as download-only (no inline rendering in the browser).

5. Supplementary Specifications

5.1 Data Model (Conceptual)

Core Entities:

- **User:** id, kaist_email, display_name, created_at
- **Course:** id, course_code, course_name, semester, klms_course_id
- **Enrollment:** user_id, course_id, verified_at, valid_until, status (active / revoked / expired)
- **Board:** id, course_id, intro_text, created_at
- **Post:** id, board_id, author_id, title, body, category, created_at
- **Comment:** id, post_id, author_id, body, created_at
- **Attachment:** id, post_id, file_name, file_url, file_size, file_extension, storage_backend, uploaded_at, expires_at
- **Assignment:** id, course_id, title, due_date, description, source (klms_synced / manual), klms_submission_status, user_completion_status
- **PersonalEvent:** id, user_id, title, start_time, end_time, description
- **MeetingEvent:** id, course_id, creator_id, title, description, date_range_start, date_range_end, time_range_start, time_range_end, finalized_start_time, finalized_end_time, status (open / finalized)
- **MeetingParticipant:** meeting_event_id, user_id, joined_at
- **MeetingAvailability:** meeting_event_id, user_id, available_slots (JSON)
- **Notification:** id, user_id, type, content, delivery_channel, target_type, target_id, target_url, is_read, created_at
- **NotificationPreference:** user_id, course_id, post_comment_enabled, deadline_enabled, meeting_enabled, email_enabled, deadline_reminder_timing (JSON, e.g., ["1d", "3h"])

5.2 MoSCoW Priority Summary

Priority	Features
Must Have	Chrome extension data parsing and sync; course-based board (basic CRUD); enrollment-verified board access; assignment calendar view; meeting scheduler (basic grid + heatmap + real-time WebSocket updates); user registration with KAIST email; bilingual UI (Korean + English)
Should Have	Limited board metadata view for non-verified KAIST users; deadline notifications; meeting time auto-suggestion; notification preferences; per-course notification control
Could Have	Pre-filled busy times in scheduler using assignments and personal events; board activity stats; upvote system; post categories; weekly email digest
Won't Have (this version)	Direct KLMS grade viewing; anonymous posting; native mobile app; professor/TA-specific features; integration with external calendars (Google Calendar, etc.)

5.3 Risks and Mitigation

Risk	Impact	Mitigation
KLMS DOM structure changes after update	Extension parsing breaks	Use Moodle Web Service API where possible; implement version detection and graceful error handling
Low user adoption	Empty course boards	Focus on scheduling features as primary value; promote within KAIST student communities
SSO-related legal/policy concerns	Project halted	The extension only reads data from an already-authenticated browser session; no SSO

Risk	Impact	Mitigation
		credentials are stored or transmitted
Chrome extension store review delays	Delayed deployment	Provide sideloading instructions for initial testing; prepare store submission early

Appendix A: Glossary

Term	Definition
Heatmap	A visual representation using color intensity to show the degree of availability overlap among participants
Manifest V3	The latest Chrome extension platform version required by the Chrome Web Store
Moodle Web Service API	A set of RESTful/SOAP APIs provided by Moodle for external systems to interact with Moodle data
DOM Parsing	Extracting structured data from the HTML structure of a web page

Appendix B: Revision History

Version	Date	Description
1.0	2026-03-31	Initial SRS draft
1.1	2026-04-03	Updated data model consistency, clarified extension role, and advanced some properties and designs